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Course on Advanced Computer Architectures

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EX1	(5 points)	
EX2	(5 points)	
EX3	(6 points)	
Q1	(5 points)	
Q2	(5 points)	
SIX QUESTIONS	(6 points)	
TOTAL	(32 points)	

EXERCISE 1 – SCOREBOARD (5 points)

Please consider the program in the table to be executed on a CPU with dynamic scheduling based on SCOREBOARD BASIC SCHEME with:

- 2 LOAD/STORE units (LDU1, LDU2) with latency 4 cycles
- 1 FP UNIT (FPU1) with latency 3 cycles
- 1 ALU/BR UNIT (ALU1) with latency 1 cycle

1. Please complete the SCOREBOARD TABLE by assuming all cache HITS

ISTRUZIONE	ISSUE	READ OPS	EXEC COMPL	WRITE BACK	Hazards Type	UNIT
I0 lw \$f5,0(\$r1)						
I1 lw \$f6,0(\$r1)						
I2 fadd \$f6,\$f6,\$f5						
I3 sw \$f6,0(\$r1)						
I4 lw \$f6,4(\$r1)						
I5 fadd \$f6,\$f6,\$f6						
I6 sw \$f6,4(\$r1)						
I7 addi \$r1,\$r1,8						

2. Express the **formula** then compute the following metrics:

- CPI = _____
- IPC = _____

EXERCISE 2 – SCOREBOARD (5 points)

Please consider the previous exercise with the introduction of the following **OPTIMIZATIONS**:

- **CHECK FOR WAW postponed to WRITE BACK phase**
- **FORWARDING PATHS**

1. Please complete the SCOREBOARD TABLE by assuming all cache HITS

ISTRUZIONE	ISSUE	READ OPS	EXEC COMPL	WRITE BACK	Hazards Type	UNIT
I0 lw \$f5,0(\$r1)						
I1 lw \$f6,0(\$r1)						
I2 fadd \$f6,\$f6,\$f5						
I3 sw \$f6,0(\$r1)						
I4 lw \$f6,4(\$r1)						
I5 fadd \$f6,\$f6,\$f6						
I6 sw \$f6,4(\$r1)						
I7 addi \$r1,\$r1,8						

2. Express the **formula** then compute the following metrics:

- **CPI** = _____
- **IPC** = _____

3. Calculate the **speedup** with respect to the SCOREBOARD BASIC SCHEME used in EXERCISE 1:

SPEEDUP = _____

EXERCISE 3: CACHE COHERENCY (6 points)

Consider the following access pattern on a two-processor system with a direct-mapped, write-back cache with one cache block and a two cache block memory.

Assume the **MESI protocol** is used, with **write-back** caches, **write-allocate**, and **write-invalidate** of other caches.

Cycle	After Operation	P0 cache block state	P1 cache block state	Memory at block 0 up to date?	Memory at block 1 up to date?
0	P0: read block 1	Exclusive (1)	Invalid	Yes	Yes
1	P1: read block 0				
2	P0: write block 0				
3	P0: write block 1				
4	P1: read block 1				
5	P1: write block 1				
6	P0: read block 0				
7	P0: write block 0				
8	P1: read block 0				
9	P0: write block 1				
10	P1: write block 1				
11	P0: read block 0				
12	P1: write block 1				
13	P1: read block 1				
14	P0: read block 1				
15	P1: write block 1				

1. Explain the main concepts of **Directory-based Protocols** to maintain the cache coherency in multiprocessor architectures (MIMD).

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QUESTION 2: (5 points)

1. Show the **Average Memory Access Time (AMAT)** in memory systems with **one level cache**.

2. Present the possible sources of cache misses (classification)

3. What is the purpose of a **second-level cache**? Show how the **Average Memory Access Time** changes in presence of the second level cache.

SIX QUESTIONS: (6 points)

Answer with **True** or **False** to the following questions:

Q1) *SIMD architectures can exploit significant data parallelism in a single operation*

Answer: T F

Q2) *In Vector Architectures, it is correct to say that: “With Operation Chaining, a vector operation can start as soon as the individual elements of its vector source operand become available”*

Answer: T F

Q3) *Multithreading aims at speeding up the execution time of each individual thread*

Answer: T F

Q4) *When using Explicit Register Renaming the ROB is no longer necessary but it can still be used to achieve precise exceptions in an easier way.*

Answer: T F

Mark the **True** answers to the following questions (some questions might have **multiple True answers**):

Q5) *Which complications are introduced to MIPS pipelined processors by adding out-of-order execution and out-of-order completion?*

Answer 1: WAR and WAW hazards **T**

Answer 2: RAW hazards **T**

Answer 3: Interrupt/Exception handling **T**

Answer 4: Control hazards **T**

Answer 5: Instruction Reordering **T**

Q6) *What are the advantages of the introduction of the Reorder Buffer?*

Answer 1: HW support for precise interrupt/exception handling **T**

Answer 2: Instructions can be executed speculatively **T**

Answer 3: Out-of-order completion of instructions **T**